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Numerical simulations of multi-hop jumping on superhydrophobic surfaces

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ABSTRACT

Water vapor condensation on superhydrophobic surfaces has received much attention in recent years because of its ability to shed water droplets via coalescence-induced droplet jumping. However, the efficient removal of jumping droplets can be limited by droplets return to the surface due to gravity, entrainment in bulk convective vapor flow, and entrainment in local condensing vapor flow. The return droplet merged with others and induced multi-hop jumping, whose mechanism still remains poorly understood. In this work, a VOF simulation is carried out to investigate the multi-hop jumping. If the returned droplets contact the droplets on the surface before touching the surface, the returned droplet coalesce with the droplets on the surface in the air, which is not conducive to shrinkage and rebounding, and the intrinsic multi-hop jumping process is inhibited. The momentum of the returned droplets is transformed into the momentum of the jumping droplets, and the multi-hop jumping process is strengthened. The multi-hop jumping velocity is linearly related to the velocity of the returned droplet. When the droplet radius ratio is less than 0.615, the momentum generated by the intrinsic multi-hop jumping can be neglected, and the multi-hop jumping velocity can be predicted base on the velocity of the returned droplet. This work will provide effective guidelines for the design of functional SHSs with enhanced droplet jumping for a wide range of industrial applications.

1. Introduction

The self-propelled jumping is widely used in micro-flow control [1–3], enhanced heat transfer [4–7], chip cooling [8–10], self-cleaning surfaces [11–15] and other areas. However, the efficient removal of jumping droplets can be limited by droplet return to the surface due to gravitational force, entrainment in the local condensing vapor flow toward the surface, etc. Systematic research on droplet jumping has been conducted through experimental observations [16–19], theoretical analysis [20–23], and numerical simulations [24–27], the well-established understanding of droplet jumping, while little attention has been paid to the motion of droplets after they detached from the surface. Most jumped droplets will return to the surface if there is no special treatment after droplet jumping off from the surface, in this case, applications based on droplet jumping will be inefficient or ineffective. Therefore, it is necessary to further study the motion process of return droplet to improve the application value of droplet jumping.

After the droplet detached from the surface, there is much force acting on it. Early studies simplified the movement of the jumped droplets as a deceleration motion under gravity and discussed the relationship between the maximum jumping height of droplets and the radius of droplets [28,29]. In the length scale of self-propelled jumping, the Bond number ($Bo < 0.1$, $Bo = \rho g R^2/\gamma$, ρ, R and γ is density, radius, and surface tension coefficient, respectively), the strength of gravity is very weak. Miljkovic et al. [30,31] has studied the process of droplet condensation and droplet jumping, point out that even if the droplet jumping direction is the same as gravity direction, the droplet will still return to the surface; The main motion resistance of droplets after detachment from the surface is the condensing vapor flow and other jumping droplets. Further, in order to overcome the resistance of condensing vapor flow, the droplet jumping is strengthened by an electric field, so that the droplets will still accelerate after detachment from the surface, and the droplet jumping can be better applied in enhanced condensation and chip cooling when the electric field is strengthened [8,10,31]. In addition to the condensing vapor flow, the convective vapor flow parallel to the surface also affect the motion of jumping droplet. There are two main types of forces exerted by the convective vapor flow on droplets; one is the drag force exerted by convective vapor flow directly on droplets, the other is the Saffman force generated by the

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Nomenclature

P	momentum (J)	ρ	water density (kg/m^3)
F	force (N)	μ	viscosity (Pa·s)
g	acceleration of gravity (m/s^2)	κ	surface curvature (1/m)
m	mass (kg)	θ	contact angle angle (°)
k	radii ratio	V_{cl}	contact line velocity (m/s)
v_0	impact velocity (m/s)	$\mathbf{v}$	velocity vector
v_d	multi-hop jumping velocity (m/s)	α	volume fraction
p	pressure (Pa)	γ	surface tension coefficient (N/m)
R_0	return droplet radius (m)	t	time (s)

velocity gradient near the wall. Birbarah et al. [32,33] theoretically analyzed the jumping process of droplets on a flat plate and inside a circular tube, under the action of convective vapor flow, the total distance traveled by jumping along the surface ranged from millimeters to meters, droplets can be detached from surface efficiently. Whether applied electric field, convection or without any treatment, the jumping droplets will still return to the surface more or less, and the droplets returning to the surface may coalesce with the droplets on the surface and induced jumping again, i.e., multi-hop jumping. Kim et al. [34] consider that the reverse momentum of the returned droplet reduces the momentum of the multi-hop jumping droplet, and then the jumping velocity of the multi-hop jumping droplet is smaller than that of the multi-droplet jumping, the momentum is not conserved under the multi-hop jumping mechanism proposed by Kim. In the subsequent theoretical analysis, it is considered that the momentum of the returned droplets will be entirely transformed into the jumping momentum of the multi-hop jumping droplets, based on which the velocity of the multi-hop jumping is predicted [32,33].

The numerical simulation is one of the main methods to study droplet jumping because it can easily obtain information about internal flow and force. In addition to different research contents, different numerical methods and boundary conditions have been used in previous work. The main model used in the simulation of droplet jumping are: volume of fluid (VOF) method [6,26,35–37], lattice Boltzmann (LB) method [25,28,29,38], Diffuse interface method [24,39], moment of fluid (MOF) method [40], level contour reconstruction method [41]. Among them, VOF and LB are commonly used methods, and each of them has its advantages and disadvantages. For the simulation of droplet jumping, three main points need to be considered, one is the accuracy of interface capturing, the other is density ratio, and the third is the accuracy of wetting representation. Both LB and VOF mark the interface by phase fraction, so the accuracy of interface capturing and wetting representation mainly depends on the grid resolution, and there is no obvious advantage or disadvantage of the two methods in this respect. VOF method has no requirement for density and viscosity ratios of simulated fluid, but LB usually uses smaller ratio (<1:100) [25,29,38] in order to calculation stability, which is deviated from the actual physical parameters (1:1000). The driving force of droplet jumping comes from the interaction between the gas-liquid interface and the solid surface. Therefore, the setting of boundary conditions on the bottom surface is the most complex and essential, and the dynamic wetting process should be fully considered [35]. In most previous studies of droplet jumping, the static contact angle is used in the wetting model. The drawback of this way is that the contact angle hysteresis is not considered, which means that there is no adhesion force between the droplet and surface, and it cannot well represent the dynamic wetting process between the droplet and the surface.

The complex motion process of returned droplets can be divided into two stages; one is a movement of droplets in the air, the other is the impact on surface or other droplets on the surface. The mechanism of the movement in the air is quite clear due to there are many studies on it, the movement of droplets can be predicted by theory, whereas the mechanism of the multi-hop jumping process induced by returned droplet impacting is lack more indepth understanding, its mechanism needs further study. Since the experiment operation of drop jumping is complicated, while it is numerical simulation method is mature, this paper used the VOF method with the dynamic contact angle model to simulate multi-hop jumping on a superhydrophobic surface. The mechanism of multi-hop jumping is analyzed, the relationships among multi-hop jumping, intrinsic multi-hop jumping, and multi-droplet jumping are clarified, which can explain in more detail the conclusions obtained from experiments in previous studies [34]. Furthermore, the effects of impact velocity, gap and droplet radius ratio on multi-hop jumping are discussed.

2. Numerical model

2.1. Numerical model

The numerical simulations used the interFoam solver in OpenFOAM which is based on the VOF [42]. The governing equations for the continuity and momentum equations are:

$$\nabla \cdot \mathbf{v} = 0 \tag{1}$$

$$\rho\left(\frac{\partial \mathbf{v}}{\partial t} + \mathbf{v} \cdot \nabla \mathbf{v}\right) = -\nabla p + \nabla\left(\mu\left(\nabla \cdot \mathbf{v} + \nabla \cdot \mathbf{v}^{\boldsymbol{T}}\right)\right) + \rho g + \boldsymbol{F}_{\mathrm{SF}} \tag{2}$$

where $\mathbf{v}$ is the velocity vector, t is the time, p is the pressure, and $\boldsymbol{F}_{\mathrm{SF}}$ is the source term generated by the surface tension. ρ and μ represent the average density and dynamic viscosity of the fluid in the cell which are functions of the liquid volume fraction in the cell, α.

$$\rho = \alpha\rho_{\mathrm{water}} + (1-\alpha)\rho_{\mathrm{air}} \tag{3}$$

$$\mu = \alpha\mu_{\mathrm{water}} + (1-\alpha)\mu_{\mathrm{air}} \tag{4}$$

The volumetric force term generated by the surface tension is calculated using the continuum surface force model (CSF):

$$\boldsymbol{F}_{\mathrm{SF}} = \gamma\kappa\nabla\alpha \tag{5}$$

where γ is the surface tension coefficient and κ is the mean curvature of the free surface:

$$\kappa = -\nabla \cdot \left(\frac{\nabla\alpha}{|\nabla\alpha|}\right) \tag{6}$$

2.2. Characteristic quantities and computational domain

The mass-averaged velocity of the droplet is defined as the droplet jumping velocity [35]:

$$v(t) = \frac{\int_\Theta \alpha v \mathrm{d}\Theta}{\int_\Theta \alpha \mathrm{d}\Theta} \tag{7}$$

where Θ represents the computational domain and v are the velocity components in the vertical directions.

In this study, in order to better represent the droplet-surface interaction, the wettability of the surface was accounted for by using the dynamic contact angle model of Kistler.

$$\theta_d = f_H\left[Ca + f_H^{-1}(\theta_S)\right] \tag{8}$$

where Ca ($Ca = \mu|V_{cl}|/\gamma$, V_{cl} is the contact line velocity between the solid and liquid) represents the capillary number. The equilibrium contact angle θ_S can be replaced by either the advancing contact angle θ_A or the receding θ_R depending on the direction of the velocity vector at the contact line, f_H is the Hoffman's function and is defined as

$$f_H = \cos^{-1}\left\{1 - 2\tanh\left[5.16\left(\frac{x}{1+1.31x^{0.99}}\right)^{0.706}\right]\right\},\ x = Ca + f_H^{-1}(\theta_S) \tag{9}$$

On an ideal smooth-surface, the momentum of a droplet is conserved on the plane (XY plane) parallel to surface, the velocity of the droplet in the XY plane can be obtained directly from the momentum conservation after impacting [32,33]. Therefore, in this paper, the velocity component in the XY plane is not considered, that is, when the droplet returns, it is perpendicular to the XY plane. In the actual condensation process, the multi-droplet coalescence (the number of the droplet is more than 2) is more common, so this paper chooses to simulate the multi-hop jumping process with three droplets involved [25,26]. The computational domain size of the computational domain was $2 \times 1 \times 1$ mm was used for the 3D simulation of multi-hop jumping as shown in Fig. 1, there are two unconnected droplets with d spacing on the superhydrophobic surface, and another droplet with a same radius of R_0 returns from above the midpoint of the droplet centroid connection at a velocity of v_0, the three droplets contact and coalesce and induced multi-hop jumping. Although such an ideal setting ignores the case that droplets may return the surface from multiple locations in actual situations, the relationships among multi-hop jumping, intrinsic multi-hop jumping, and multi-droplet jumping can still be explained from the trend. The advancing contact angle θ_A and receding contact angle θ_R of superhydrophobic surfaces are 162°and 160° respectively. To ensure that the results were independent of the domain resolution, three different sized meshes with $200 \times 100 \times 100$ elements, $300 \times 150 \times 150$ elements, and $400 \times 225 \times 225$ elements were used to simulate the multi-hop jumping ($R_0 = 100\ \mu$m, $v_0 = 1.0$ m/s, $d = 100\ \mu$m). The jumping velocities obtained from simulation were 0.2719 m/s, 0.2582 m/s and 0.2507 m/s for the three meshes. When the grid resolution was increased from $300 \times 150 \times 150$ elements to $400 \times 225 \times 225$ elements, the jumping velocities increased 5.03% and 2.90% (<5.00%). Thus, the $300 \times 150 \times 150$ mesh was used for other cases. The time resolution δt is controlled by the maximum Co (Co = max(|U|)$\delta t/\delta x$, δx is the space step, max(|U|) is the largest velocity in the computation domain) number, and the maximum Co number is set to 0.5 in this paper. The superhydrophobic surface was set as a no-slip boundary with the other boundaries as pressure outlet boundaries [26,27,35]. The physical parameters are listed in Table 1. As with previous experiments and simulations, the gas and liquid physical parameters were for air and water at 20 °C [26,43].

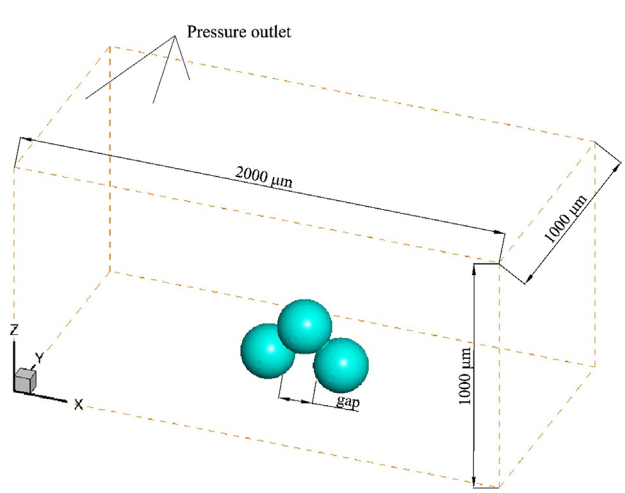

Fig. 1. 3D computational domain and boundary conditions.

Table 1
Fluid properties of air and water at 20 °C (g = 9.8 m s^{-2}).

Fluids	γ (N m^{-1})	ρ (kg m^{-3})	μ (Pa s)
Water	0.072	998.2	1.0×10^{-3}
Air		1.2	1.8×10^{-5}

3. Numerical result and discussion

3.1. Impact velocity and the gap

Multi-hop jumping can be classified into two categories according to the size of the gap. In the first category, the gap $d < 2R_0$, the returned droplet touch with others droplets fist and then touch the surface; in the second category, the gap $d > 2R_0$, the returned droplet touches the surface first, then deforms and expands to touch others droplets. For the first category, the gap d can be easily set to $0.5R_0$, $1.0R_0$, $1.5R_0$, For the second category, the maximum spreading radius should be considered, and the maximum spreading radius predicted by the existing theoretical model is about $1.4R_0$ [44,45], therefore, the gap d set to $2.1R_0$ and $2.4R_0$. The gap range of the first category is larger than that in the second category, that is, the probability of the first category is higher than that in the second category. Therefore, the discussion in the latter part is mainly about the multi-hop jumping in the first category.

If the droplet is considered only to return to the surface under gravity, its returned velocity(impact velocity) equal to the initial jumping velocity $v_{\text{initial}} = C_0\ (\gamma/\rho_{\text{water}}R_{\text{initial}})^{0.5}$. According to the number and arrangement of the initial droplets, prefactors C_0 can be 0.2–0.37 [16,25], in addition, with the enhancement of the external force field such as electric field, the value of C_0 can be larger [31]; if the effect of condensing vapor flow is considered, the impact velocity will be great than v_{initial}. The droplet terminal velocity is assuming Stokes flow past a sphere ($v_{\text{terminal}} \approx 2\rho_{\text{water}}gR_0^2/9\mu_{\text{air}}$). The radius of returned droplet discussed in this paper is fixed as $100\ \mu$m, the range of impact velocity is about 0.2–1.47 m/s or $U_0 \sim 6.85U_0$ ($U_0 = 0.2\ (\gamma/\rho_{\text{water}}R_{\text{initial}})^{0.5}$), in order to study the effect of impact on the coalescence process (hereinafter referred to as the intrinsic multi-hop jumping), a case with impact velocities of 0 m/s is set up. In summary, the impact velocity of returned droplets is set to 0, 0.2, 0.4, 0.6, 0.8, 1.0 m/s and can also written as $0U_0$, $2U_0$, $3U_0$, $4U_0$, $5U_0$ in this paper.

3.2. Model validation and droplet morphology evolution during intrinsic multi-hop jumping

The maximum value of the mass-averaged velocity during the jumping process was regarded as the jumping-velocity and used to be compared with results by other researches to validate the model. For droplet jumping induced by coalescence of three identical droplets (100 μm radii) on the superhydrophobic with a contact angle of 160°, Wang et al. theoretically calculated the jumping-velocity ad 0.281 m/s; Chu et al. obtained the jumping-velocity from their simulation ad 0.245 m/s; in the present model, the jumping-velocity is extracted to be 0.269 m/s, which is consistent with other results with an average deviation of 2.28%.

Fig. 2(a and b) shows the morphological evolution of multi-hop jumping ($v_0 = 0$ m/s, intrinsic multi-hop jumping) and multi-droplet jumping with an impact velocity of 0 m/s. As shown in Fig. 2(a), the liquid bridge develops rapidly after coalescence beginning (0.00–0.05 ms), subsequently, the droplets begin to shrink (0.05–0.4 ms) and form a reflux perpendicular to the surface, finally, under the reaction of the surface, the droplets detach from the surface (0.4–0.7 ms). Comparing with Fig. 2(a) and (b), the multi-droplet jumping will also undergo three processes: liquid bridge development (0.00–0.05 ms), shrinkage (0.05–0.35 ms), and rebounding (0.35–2.5 ms). The development direction of the liquid bridge of multi-droplet jumping is perpendicular to the surface, but the development direction of the liquid bridge of intrinsic multi-hop jumping is not perpendicular to the surface, it is more like coalescence in the air. In addition, in the process of multi-droplet jumping, the droplet shrinks mainly in the horizontal direction, while the intrinsic multi-hop jumping shrinkage process has both horizontally and vertically direction. The surface reaction force causes droplets to jump, which can be expressed by the pressure of bottom slices. As shown in Fig. 2(c), the two differences between intrinsic multi-hop jumping and multi-droplet jumping are adverse effect the jumping in two ways respectively. The first is that the coalescence of droplets is completed in the air, and droplets cannot contact the bottom (0.2 ms), the second is that the shrinkage is not concentrated in the horizontal direction, resulting in the reduction of the pressure on the bottom slice (0.05 ms and 0.4 ms), all of these reduce the reaction force. As a result, the droplets of intrinsic multi-hop jumping cannot completely detach from the surface, but oscillate near the surface, while the droplets detached the surface at a greater velocity in multi-droplet jumping. The relationship among intrinsic multi-hop jumping and multi-droplet jumping is consistent with Kim's conclusion that multi-hop coalescence resulted in reduced jumping velocity due to the out-of-plane coalescence and reduced momentum transfer for jumping [34]. Moreover, the mechanism proposed in this paper is more detailed and clearer than that by Kim, and it is the basis for later discussing.

3.3. Influence of impact velocity and the gap

Figs. 3 and 4 show the droplet morphological evolution with time for different impact velocities in the two categories mentioned in Section 3.1, respectively. As shown in Fig. 3, when the impact velocity is not zero, the droplet morphology evolution is similar to that of the intrinsic multi-hop jumping, which can be roughly divided into three stages: the development of the liquid bridge, shrinkage, and rebounding. The development of liquid bridge is dominated by the surface tension, as the Weber number (we < 2, define as we = $\rho_{\mathrm{water}} R_0 v_0/\gamma$) of the returned droplet is very small, compared with the surface tension, the inertia force is relatively weak. Therefore, although the droplets return at different velocities in the three cases, there is little difference in the droplet morphology during the liquid bridge development stage (0–0.05 ms). On the shrinkage stage (0.05–0.15 ms), the inertia force begins to affect the evolution of the droplet morphology. In the vertical direction, the middle part of the droplet is depressed under the action of the inertia force, and the depth of the depression increases with the increase of the impact velocity, and the middle part of the droplet also contacts the bottom surface earlier. In the horizontal direction, the droplets shrink and gather in the middle under the action of surface tension. On the rebounding stage (0.15–0.55 ms), because the larger the impact velocity is, the earlier the middle of the droplet reaches the surface and the earlier the rebound time is, therefore, the closer the top of the droplet to the shape of the bullet and the earlier it detaches from the surface as the impact velocity increases.

As shown in Fig. 4, in the second category, the morphological evolution of the droplets can also be divided into three stages: the development of the liquid bridge, shrinkage, and rebounding. During the liquid bridge development stage (0–0.05 ms), unlike the multi-hop jumping in the first category, the shape of liquid bridge varies significantly for different impact velocities because the returned droplets first contact the surface and deform and then coalesce with the other two droplets. When the impact velocity is U_0, the liquid bridges are almost parallel to the surface, while the impact velocity is $3U_0$, $5U_0$, the liquid bridge is not parallel to the surface. On the early stage of shrinkage (0.05–0.10 ms), as in the first category, the center of the droplet is depressed under the action of inertial force in the vertical direction, and the higher the impact velocity, the deeper the depression. At the rebounding stage, the morphological evolution of droplets is utterly different from that in the first category (0.10–0.25 ms); the fluid in the middle of the droplet begins to rebound under the surface reaction force, this rebound makes the middle of the droplet detach from the surface, the higher the impact velocity, the higher the rebound height. Because the rebound force in the middle of the droplet is insufficient to push the whole droplet to detach from the surface, the liquid in the middle of the droplet is dragged back to the surface through the surface tension by the remaining liquid attaching on the bottom surface, marking the end of the shrinkage process, this phenomenon is marked "hindrance process".

Fig. 5(a) and (b) are the evolution of the centroid velocity of the droplet with time under different impact velocities when the gap is $1.5R_0$ and $2.4R_0$, respectively. As shown in Fig. 5(a), the evolution trend of velocity is very similar to that in the first category, in the acceleration stage, the droplet accelerates with a relatively stable acceleration. From the previous discussion, it can be seen that if the returned droplets coalesce with other droplets before contacting the surface, their morphological evolution are similar to the multi-droplet jumping, so their velocity evolution is similar. With the increase of impact velocity, the jumping droplets get more momentum, so the jumping velocity is also higher. When the impact velocity is small (U_0, $3U_0$), the multi-hop jumping velocity is less than the multi-droplet jumping velocity, and when the impact speed is large enough ($5U_0$), the multi-hop jumping velocity even exceeds the multi-droplet jumping velocity. As shown in Fig. 5(b), in the second category, when the impact velocity is small (U_0), the evolution trend of the droplet velocity is similar to that of the multi-droplet jumping; when the impact velocity is large ($3U_0$, $5U_0$,), the evolution trend of the velocity obviously changes, and the acceleration process consists of two distinctly different stages. As mentioned in the previous discussion of the morphological evolution of multi-hop jumping droplet in the second category, there is a hindrance process in the process of droplet shrinkage, which destroys the jumping process, and the acceleration of the droplet decreases suddenly. After the hindrance process, the acceleration becomes stable again.

The momentum conversion model of multi-hop jumping is imperfect in previous studies; it assumes that the impact process

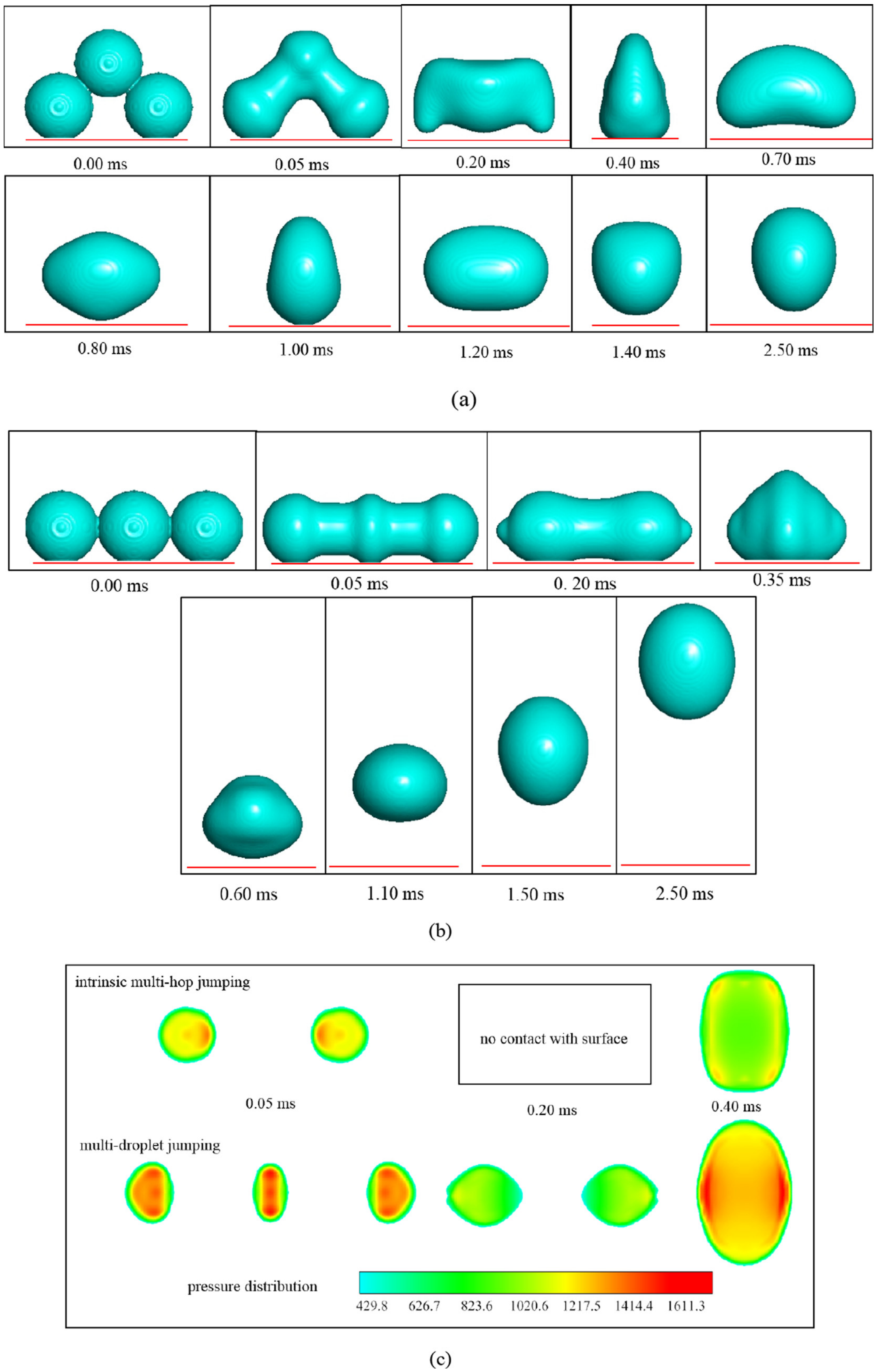

Fig. 2. Droplet morphological evolution in the XZ plane and pressure distribution on bottom slice, the red line is surface. (a) Intrinsic multi-hop jumping, droplet radius R_0 is 100 μm, gap $d = R_0$ is 100 μm, impact velocity is 0.0 m/s; (b) multi-droplet jumping, droplet radius R_0 is 100 μm, gap $d = 2R_0$ is 200 μm, impact velocity is 0.0 m/s; (c) pressure distribution on bottom slice, the upper and lower corresponding to the case in a and b respectively. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

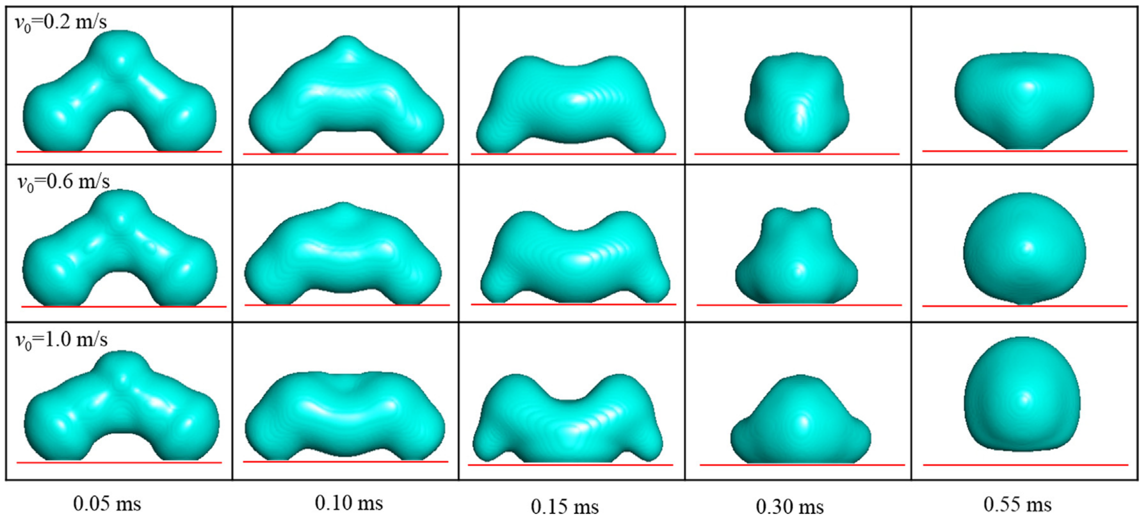

Fig. 3. The droplet morphological evolution in the XZ plane with different impact velocities in the first category. From top to bottom, the impact velocities are U_0, $3U_0$, $5U_0$, respectively. The droplet morphology at zero time is the same as that at Fig. 3(a) 0.00 ms, and the radius of the three droplets is 100 μm, the gap $d = 1.5R_0$ is 100 μm.

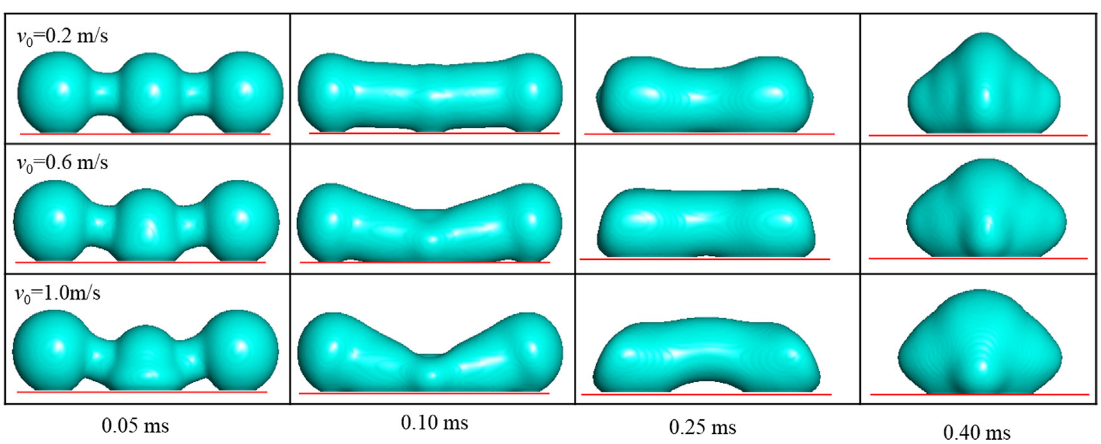

Fig. 4. The droplet morphology evolution in the XZ plane with different impact velocities in the second category. From top to bottom, the impact velocities are U_0, $3U_0$, $5U_0$, respectively. The droplet morphology at zero time is the same as that at Fig. 3(a) 0.00 ms, and the radius R_0 of the three droplets is 100 μm, the gap $d = 2.1R_0$ is 210 μm.

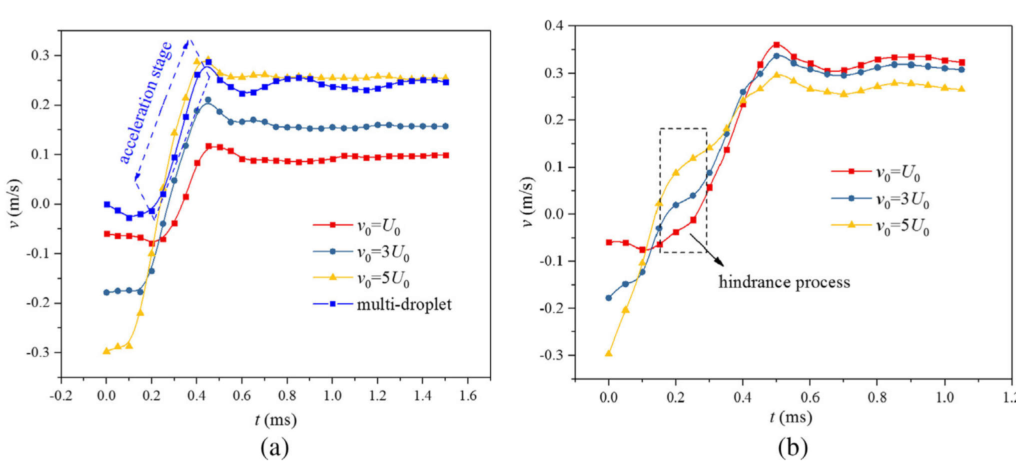

Fig. 5. The velocity components in the Z-direction of droplets with different impact velocities vary with time. From top to bottom, the droplet impact velocities are U_0, $3U_0$, $5U_0$, and the gaps are $1.5R_0$ and $2.4R_0$, respectively.

does not affect the droplet jumping and that the momentum carried by the returned droplet can be entirely converted into the momentum of the multi-hop jumping. Therefore, an improved model was proposed, the momentum of droplets in multi-hop jumping comes from the momentum generated by the intrinsic multi-hop jumping and the momentum carried by the returned droplets:

$$P_d = C_1 P_{multi\text{-}droplet} - C_2 P_{impact} \tag{10}$$

And the multi-hop jumping velocity can be obtained as follows:

$$v_{dt} = C_1 v_{multidrop} - \frac{m_1}{(m_1 + 2m_2)} C_2 v_0 \tag{11}$$

where P_d is the momentum of multi-hop jumping, $P_{multi\text{-}droplet}$ is the momentum of multi-droplet jumping, P_{impact} is the momentum carried by the returned droplets, v_{dt} is the multi-hop jumping velocity, $v_{multi\text{-}droplet}$ is the multi-droplet jumping velocity. In Eq. (10), the first term on the right side is the momentum generated by the intrinsic multi-hop jumping, and the coefficients C_1 is used to characterize the adverse effect of intrinsic multi-hop coalesces process on the multi-droplet jumping; the second term is the momentum provided by the returned droplets; and C_2 represents the ratio of the momentum carried by the returned droplets to that converted to the jumping momentum, the ratio is related to we number and viscosity, in all cases discussed in this paper, the range of We is 0–1, it can be regarded as constant and viscosity unchanged, so C_2 is constant, the mean value of the case discussed in this paper is about 0.853 ± 0.008 [46]. Fig. 6 is the relationship between the multi-hop jumping velocity and the impact velocity under the different gap. As shown in the figure, when the gap is less than $2R_0$, it is the first category of multi-hop jumping. The multi-hop jumping velocity increases linearly with the increase of impact velocity and the three lines with a gap of $0.5R_0$, $1.0R_0$, and $1.5R_0$ are almost parallel, so C_1 is constant when the gap is the same, that is to say, the change of impact velocity does not affect the momentum generated by intrinsic multi-hop jumping. From the discussion in Section 3.1, it can be seen that the smaller the gap, the more unfavorable the multi-hop jumping. Therefore, with the increase of the gap, the momentum generation process of intrinsic multi-hop jumping becomes more similar to the momentum generation process of multi-droplet jumping, so that the C_1 increases as well as the multi-hop jumping velocity of droplets also increases. In this paper, the corresponding C_1 for gap $0.5R_0$, $1.0R_0$ and $1.5R_0$ are 0, 0.24, 0.4 and 0.84 respectively. When the gap is greater than $2R_0$, although the increase of impact velocity brings more momentum, because the hindrance process on the droplet shrinkage stage, the momentum generated by intrinsic multi-hop jumping decreases, that is, C_1 decreases. Therefore, in the second category, the multi-hop jumping velocity no longer increases linearly with the increase of impact velocity, but increases first, and decrease. With the increase of the gap, the hindering effect becomes stronger, the greater the amplitude of the decrease of the multi-hop jumping velocity is.

3.4. Influence of the radii ratio

In an actual condensation process, the coalescence of droplets with different radius is more common. Assuming that the radius ratio of the returned droplet to the droplet on the surface is $k = R_0/R_1$, the Eq. (11) can be written:

$$v_{dt} = C_1 v_{multidrop} - \frac{1}{1 + 2k^3(4 - (2 - cos\theta)(1 + cos\theta)^2)} C_2 v_0 \tag{12}$$

The smaller the radius ratio is, the lower the energy conversion efficiency (the ratio of the released surface energy induced by coalescence to that converted to jumping kinetic energy) is [38], therefore, the $v_{multi\text{-}droplet}$ decreases with the decrease of k. When k is less than a certain value, the jumping will not occur, that is, $v_{multi\text{-}droplet} = 0$, in this case, the momentum of the multi-hop jumping is entirely provided by the returned droplet, and the multi-hop jumping velocity can be expressed as

$$v_{dt} = -\frac{1}{1 + 2k^3(4 - (2 - cos\theta)(1 + cos\theta)^2)} C_2 v_0 \tag{13}$$

Fig. 7 is the relative difference ($\varepsilon = \text{abs}\ (v_d - v_{dt})/v_{dt}$) between the theoretical velocity predicted by (13) and the velocity obtained from the simulation for different radius ratios. As shown in the figure, the relative difference between the theoretical velocity and the simulation velocity decreases rapidly with the decrease of the radius ratio, and the relative difference between the predicted velocity and the actual velocity by formula (13) is less than 10% when the radius ratio less than 0.615. Therefore, it can be considered that the momentum generated by the intrinsic multi-hop jumping can be ignored when the radius ratio k less than 0.615, and the velocity of multi-hop jumping can be predicted directly by Eq. (13).

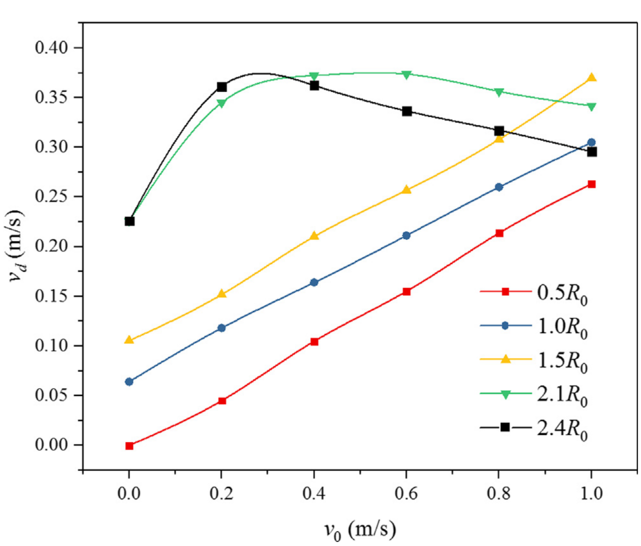

Fig. 6. The relationship between the multi-hop jumping velocity and the impact velocity under the different gap. The radius R_0 of the three droplets is 100 μm.

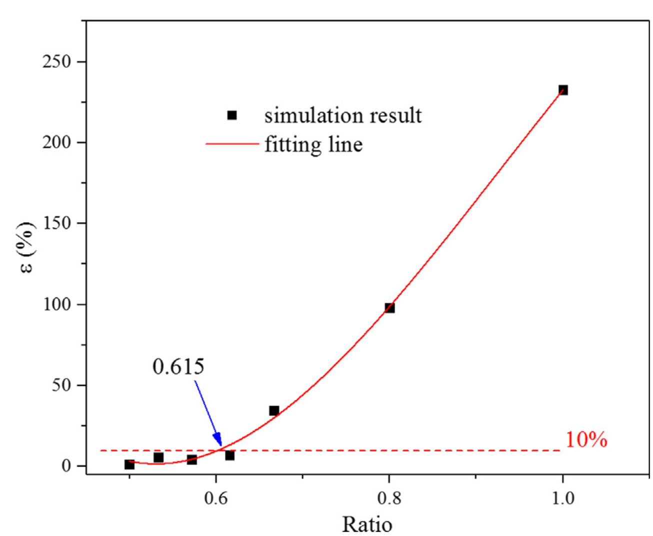

Fig. 7. The relative difference between the theoretical velocity and the velocity obtained from the simulation for different radius ratios.

4. Conclusions

The VOF method was used to simulate the multi-hop jumping of droplets on a superhydrophobic surface. The mechanism of multi-hop jumping is analyzed, and the effect of the gap and the returning impact velocity on multi-hop jumping is discussed. The main conclusions are:

(1) During the coalescences process of multi-hop jumping, the development direction of the liquid bridge is not perpendicular to the surface, which is adverse to generated jumping momentum, and this effect decreases with the increase of gap.

(2) When the gap is less than $2R_0$, the impact velocity does not affect the process of jumping momentum generated by intrinsic multi-hop jumping, and the ratio of momentum carried by the returned droplets to that converted to multi-hop jumping momentum is almost constant. When the gap is fixed, the velocity of multi-hop jumping increases with the increase of droplet impact velocity. When the gap is larger than $2R_0$, the hindrance process in the droplet shrinkage stage destroys the process that the intrinsic multi-hop jumping generates jumping momentum. The higher the impact velocity, the stronger the hindering effect. The velocity of multi-hop jumping increases first and then decreases with the increase of impact velocity.

(3) The momentum generated by the intrinsic multi-hop jumping decreases as radius ratio decreases, when the radius ratio of droplets is less than a certain value, the jumping momentum generated by intrinsic multi-hop jumping can be neglected. The multi-hop jumping velocity is only related to the momentum of the returned droplets; the critical radius ratio is about 0.615.

In this paper, the relationship among multi-hop jumping, intrinsic multi-hop jumping, and multi-droplet jumping are discussed under several ideal conditions. However, in the actual process of multi-hop jumping, the impact position and angle of the droplet should be considered, this case is also fascinating and worth studying in future work.

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Conflict of interest statement

We declare no any interest conflict.

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